

Ecology and Biodiversity Assessment of IIT Kanpur campus



A Biodiversity and Carbon Sequestration study was conducted in the premises of Indian Institute of Technology (IIT) Kanpur campus, located in Uttar Pradesh, to assess the baseline biodiversity and estimate botanical carbon sequestration. The sprawling green campus is spread over an area of 1055 Acres with rich biodiversity of native and rare species. The study was carried out in the pre-monsoon season during the third week of March, 2026 with a team consisting of three zoologists and three botanists with two botany assistants, under sunny and mild conditions.

Key Project Objectives:

- Document the campus flora and fauna, and quantify species richness using standard ecological indices (Simpson's and Shannon's).
- Assess conservation statuses as per the IUCN Red List and the Wildlife Protection Act (1972).
- Identify biodiversity hotspots and recommend conservation measures, such as water body restoration and habitat improvement.
- Promote community participation by actively involving students, faculty, and staff volunteers.

Survey Highlights:

- **Scientific Methodology:** Our expert teams systematically sampled the habitats using transect walks, night trails, and ~ 60-65 botanical quadrats. We deployed high-resolution cameras and bird sound recorders to accurately capture data in dense canopies and low-light conditions.
- **Thriving Urban Ecology:** The campus supports an incredibly high ecological diversity for an urban environment.
- **Campus Hotspots:** Zones like the *Aam Bagiya* and campus wetlands were identified as biodiversity-rich areas with active water bodies supporting multiple trophic levels.
- **Community Integration:** Over 20 volunteers joined our field experts, ensuring the strong integration of local ecological knowledge into our findings.

Key Findings:

- A total of **141** flora species were observed, with **56** trees, **15** species of shrubs, **62** species of herbs, and **8** species of climbers.
 - **65.44%** of the observed flora species were found to be native to the Indian subcontinent. While more than **70%** of the tree species were native (**71.43%** native species), there were **58.62%** native herbs, **100%** native climbers and **50%** native

shrubs



- A faunal survey revealed the presence of **8** mammal species through direct observation. **75** bird species, **4** reptiles and **4** Amphibians were observed. Among insects, **65** distinct species were observed, belonging to diverse orders (**24** butterflies & moths, **5** dragonflies & damselflies, **17** spiders and **19** others like ants, flies, bees etc.).



- Simpson's and Shannon's index for avifauna indicated high biodiversity. For birds, Shannon's index value varied from 2.9742 to 3.6815, whereas Simpson's index value varied from 0.9413 - 0.9562.
- Carbon Sequestration was estimated by extrapolating quadrat-based tree density data and averaging the height and girth of trees, to in the range of 8053.5 tonne. As per Habitat Quality Index of InVEST, score is 3.3 out of 5.



This critical data successfully laid the foundation for long-term ecological stewardship and climate-resilient infrastructure planning at IIT Kanpur.